DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING • PES UNIVERSITY, BENGALURU

UE24CS395: Database Management Systems Mini Project • Academic Year 2025–2026

Entity-Relationship Modeling & Conceptual Design

Evaluation Matrix Row 1: ER Diagram • Weightage: 2 Marks

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1. Executive Summary & Domain Scope

The Intelligence Fusion Platform is an operational case-management and intelligence integration system designed for multi-agency law enforcement coordination. The relational model models cases, assigned investigative personnel, implicated subjects, commercial organizations, motor vehicles, digital devices, geospatial coordinates, physical exhibits, and custody transfers.

The schema consists of **22 normalized tables**: 10 primary entity/audit relations and 12 associative junction tables. To maintain strict visual clarity without clutter, we present our entity-relationship design in both **Chen notation** (conceptual modeling) and **Crow's-Foot notation** (physical implementation), organized as an overarching system overview and modular domain clusters.

2. Conceptual ER Modeling Notation & Semantics

In our Chen notation diagrams, we adhere to formal academic database design semantics:

Notation Element	Visual Representation	Applied Semantics in Project Schema
Strong Entity	Single Rectangle	Independent domain relations: `officers`, `cases`, `people`, `organizations`, `vehicles`, `devices`, `locations`, `evidence`.
Weak Entity	Double Rectangle	Existence-dependent audit logs: `evidence_custody` (depends on `evidence`) and `case_activity` (depends on `cases`).
Relationship	Single Diamond	Binary associations connecting participating entities with labeled cardinalities (1:1, 1:N, M:N).
Identifying Relationship	Double Diamond	Binds weak entities to owner parents with double connecting lines signifying total participation.
Attributes & Keys	Ellipses	Underlined text indicates Primary Keys; dashed underlining indicates partial discriminators.
Relationship Attributes	Ellipses on Diamonds	Attributes stored on M:N associative junction bridges (e.g. `role_in_case`, `association_type`, `added_at`).

3. Complete Conceptual ER Model (Chen Notation Overview)

Figure 1 illustrates the complete conceptual architecture across all 10 entities and their 17 primary relationships (including the 12 many-to-many junction bridges modeled as relationship diamonds).

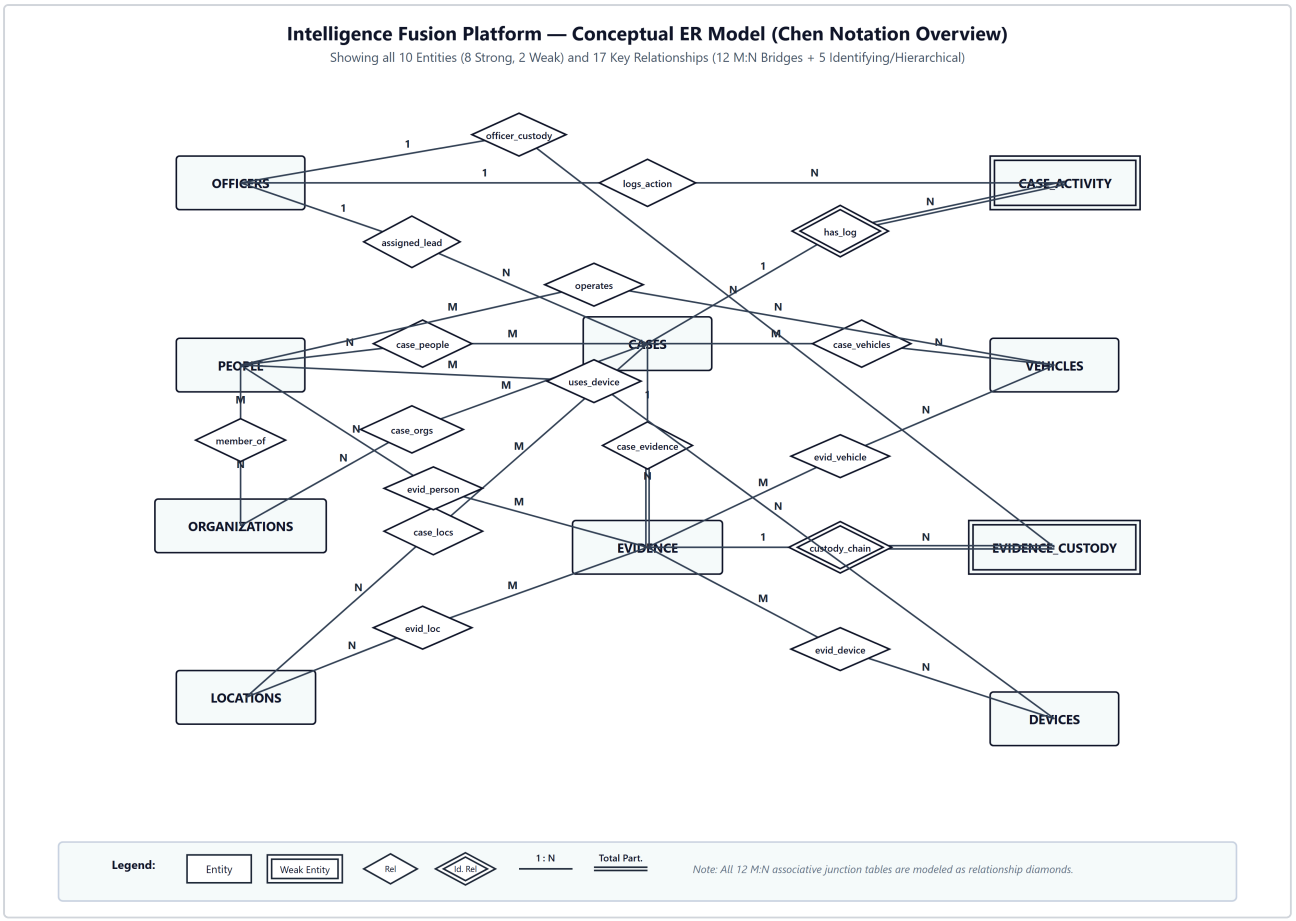


Figure 1: *Complete Conceptual ER Diagram (Chen Notation Overview). Depicts all 10 entity relations, identifying custody/audit relationships, and 12 M:N associative links.*

4. Chen Notation Domain Clusters

To ensure every attribute oval, key constraint, and relationship parameter is readable at 100% scale on A4 paper, we divide the conceptual model into four focused domain clusters:

4.1 Case Management & Automated Audit Cluster

Figure 2 models core case provisioning and immutable audit trail capture. Weak entity `case_activity` exhibits total participation in the identifying relationship `logs_activity` owned by `cases`.

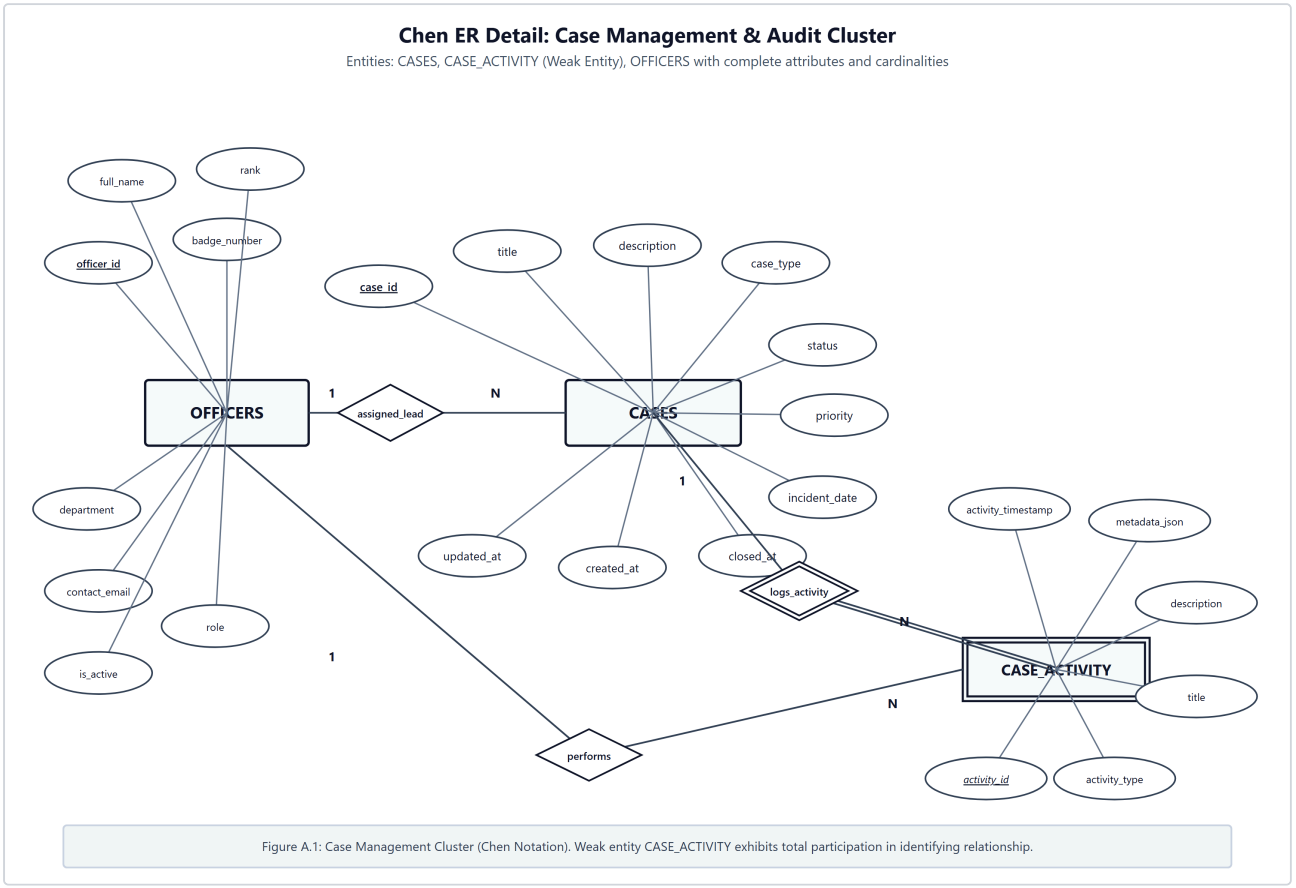


Figure 2: *Case Management Cluster (Chen Notation). Demonstrating `CASES`, weak entity `CASE_ACTIVITY`, and lead investigator assignment from `OFFICERS`.*

4.2 People, Organizations & Physical Assets Cluster

Figure 3 captures implicated actors, criminal organizations, vehicles, mobile hardware, and physical incident coordinates. Associations between entities carry specific attributes (e.g., `role` in organization, `relationship_type` for vehicles and devices).

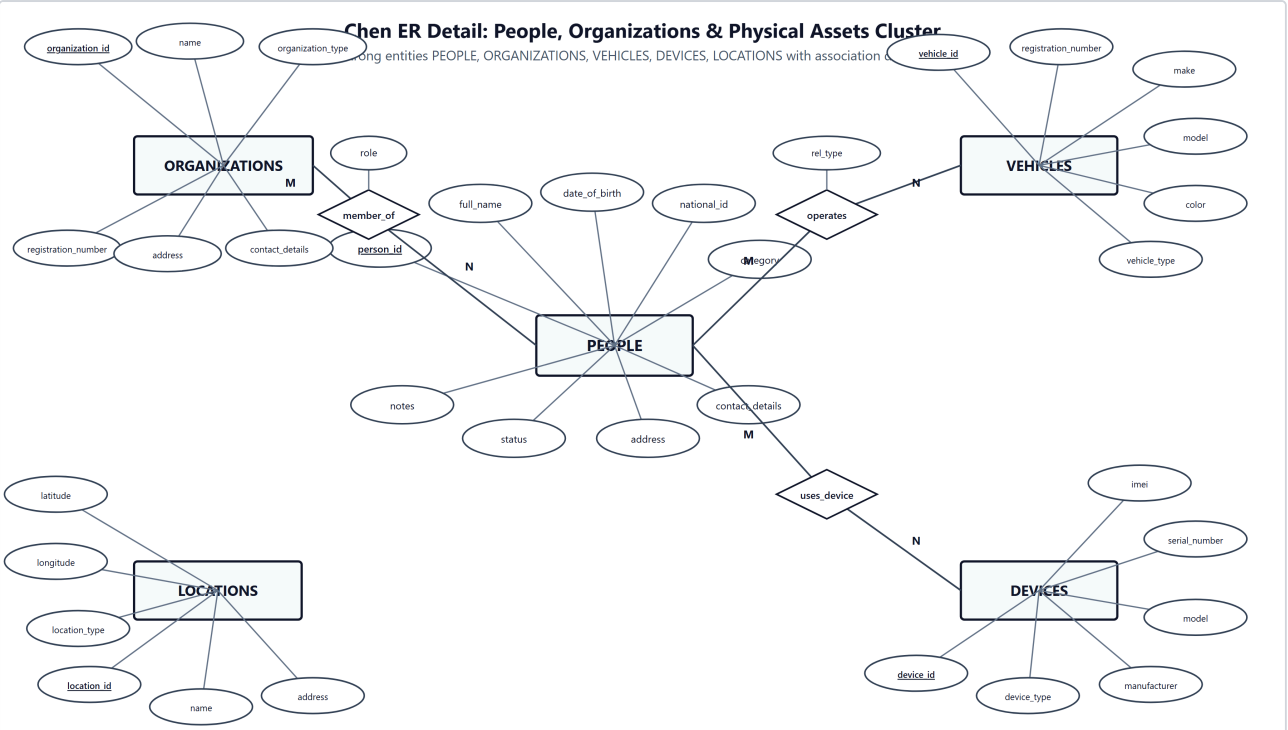


Figure A.2: People, Organizations & Physical Assets Cluster (Chen Notation). Association attributes shown on diamonds.

Figure 3: *People, Organizations & Physical Assets Cluster (Chen Notation). Models multi-party associations and asset ownership links.*

4.3 Evidence & Chain of Custody Cluster

Figure 4 details physical and digital exhibits seized during investigations. Weak entity `evidence_custody` enforces an unbroken, tamper-evident transfer record via identifying relationship `custody_chain` with parent `evidence`.

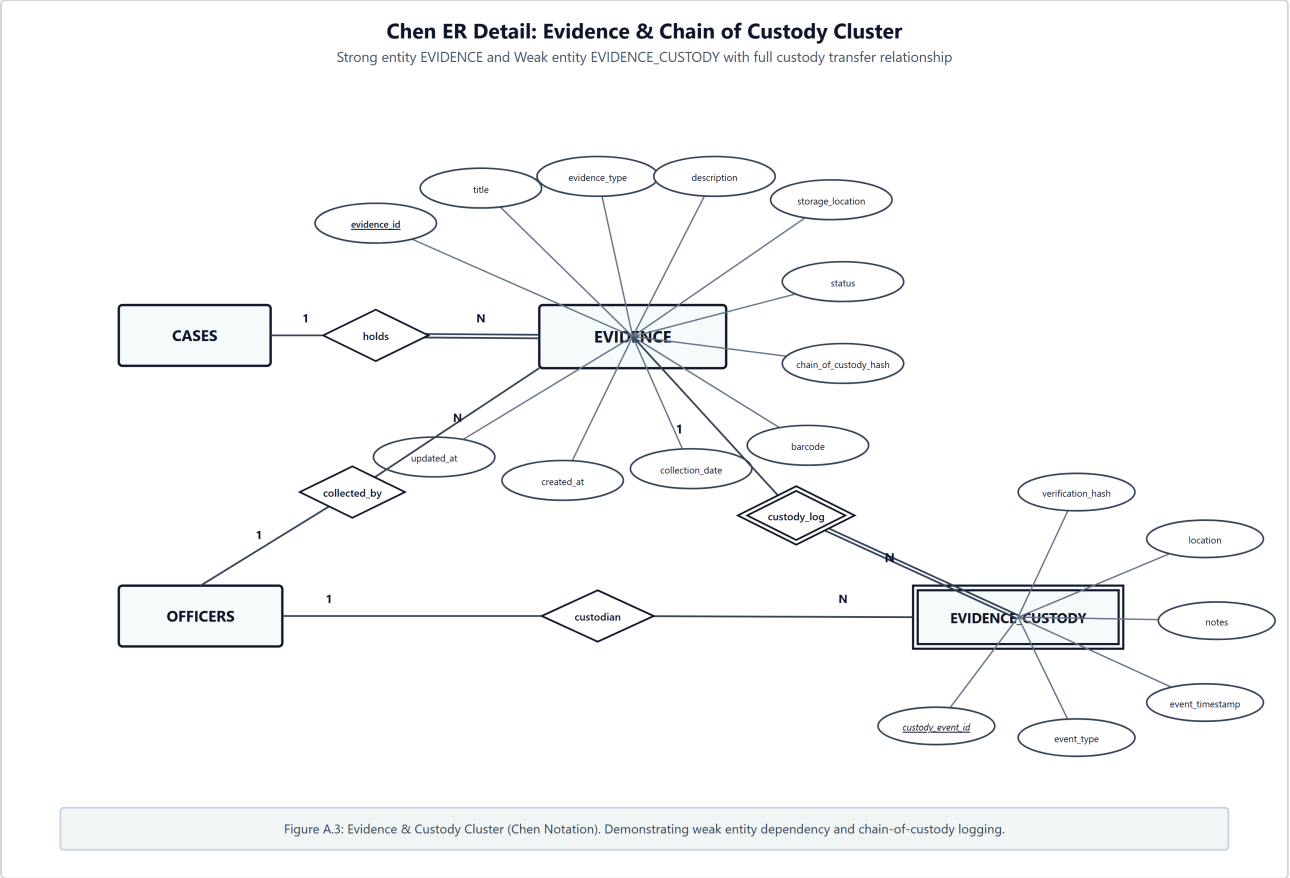


Figure 4: *Evidence & Chain of Custody Cluster (Chen Notation). Captures physical exhibits, SHA-256 integrity verification hashes, and custody event transitions.*

4.4 Officers & Audit Accountability Cluster

Figure 5 illustrates accountability linkages where law enforcement officers act as case leads, evidence collecting agents, custody transfer recipients, and system activity performers.

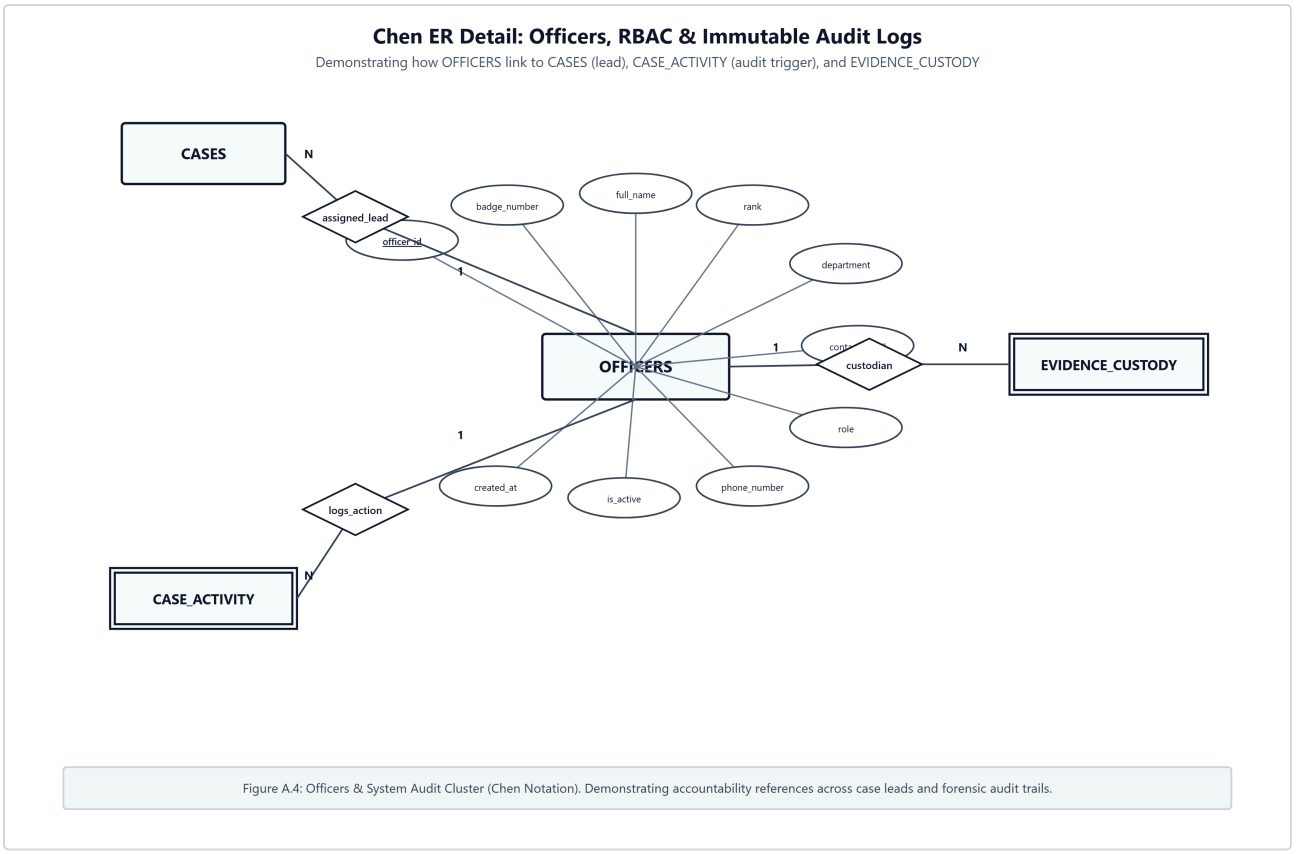


Figure 5: *Officers & Accountability Cluster (Chen Notation). Illustrates role-based operational oversight.*

5. Complete Physical Relational Schema (Crow's-Foot Notation)

Figure 6 presents the physical relational implementation of the database in Crow's-Foot notation, comprising all 22 tables. Primary keys (PK) and foreign keys (FK) are typed, and cardinalities represent exact database engine enforcement.

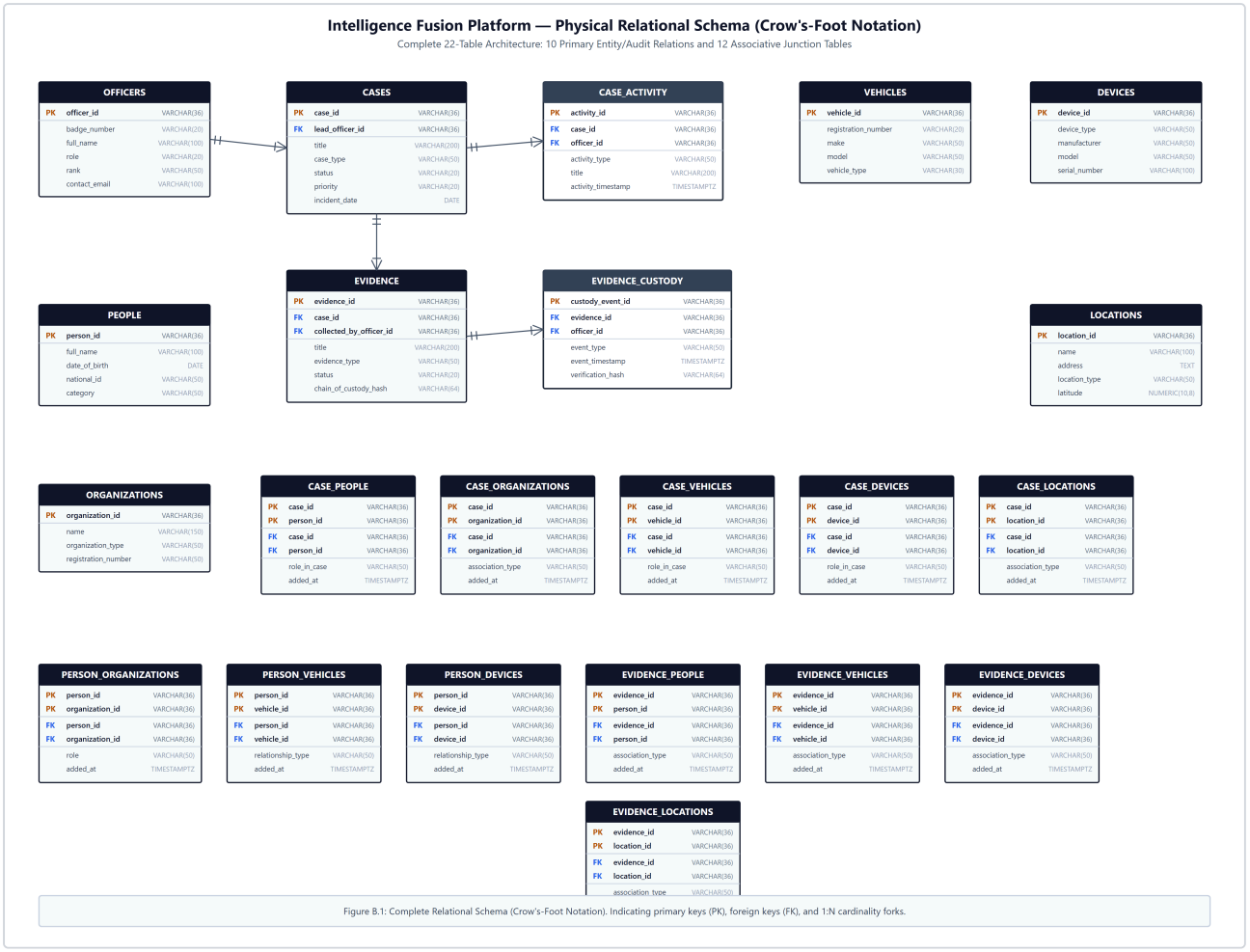


Figure 6: *Complete Physical Relational Schema (Crow's-Foot Notation). Depicting all 22 tables, column data types, foreign key constraints, and 1:N cardinality forks.*

6. Key Database Design Choices

We implemented three critical architectural decisions during data modeling:

1. **Explicit Associative Junction Tables (12 Bridges):** Rather than collapsing many-to-many relationships or storing comma-separated foreign keys in text fields (violating 1NF), we created 12 dedicated junction relations. Each carries its own composite primary key and audit metadata (`added_at`, `role_in_case`, `notes`).
2. **Weak Entities for Immutable Audit Logs:** Both `case_activity` and `evidence_custody` cannot exist without a valid parent case or evidence item. Their life cycle is bound to the parent entity via foreign key constraints with `ON DELETE CASCADE` or `RESTRICT`.
3. **Standardized Surrogate Key Scheme:** Primary keys utilize structured alphanumeric strings (e.g. `CASE-2026-001`, `PER-001`, `EVD-001`, `OFF-001`) backed by `VARCHAR(36)`. This supports human auditing, physical barcode labeling, and direct migration between PostgreSQL and SQLite.